

# Lixiang Zhao

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## SUMMARY

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My research focuses on Human-Computer Interaction and Interactive Scientific Visualization and Computer Graphics. I focus on designing and evaluating spatial interaction and visualization techniques for specific scientific domains (such as astronomy, biology, medicine, etc.) in XR environments to meet experts' needs and boost their data comprehension and exploration.

## STUDY EXPERIENCE

### University of Liverpool (UoL), United Kingdom

12/2021 – present

Ph.D. in Computer Science (Off-Site PhD Programme at Xi'an Jiaotong-Liverpool University)

Supervisors: Lingyun Yu, Yue Li, Floriana Grasso, Hai-Ning Liang and Tobias Isenberg

### Xi'an Jiaotong-Liverpool University (XJTLU), China

09/2017 – 07/2021

### University of Liverpool (UoL), United Kingdom

BEng in Telecommunication Engineering | First Class (Honours)

## PUBLICATIONS

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### Peer-reviewed Articles

- [1] **L. Zhao**, F. Xie, T. Isenberg, H.-N. Liang and L. Yu, "ScaleFree: Dynamic KDE for Multiscale Point Cloud Exploration in VR," in IEEE Conference on Virtual Reality and 3D User Interfaces, Mar. 2026, pp. 358-368, doi: [10.1109/VR67842.2026.00057](https://doi.org/10.1109/VR67842.2026.00057).
- [2] **L. Zhao**, T. Isenberg, F. Xie, H.-N. Liang and L. Yu, "SpatialTouch: Exploring Spatial Data Visualizations in Cross-reality," in IEEE Transactions on Visualization and Computer Graphics, vol. 31, no. 1, pp. 897-907, Oct.2025, doi:[10.1109/TVCG.2024.3456368](https://doi.org/10.1109/TVCG.2024.3456368).
- [3] W. Xu, **L. Zhao**, H. Song, X. Song, Z. Lu, Y. Liu, M. Chen, E. G. Lim, and L. Yu. "Mozualization: Crafting Music and Visual Representation with Multimodal AI." in CHI Conference on Human Factors in Computing Systems Extended Abstracts, Apr. 2025, doi:[10.1145/3706599.3719686](https://doi.org/10.1145/3706599.3719686).
- [4] H Yao, **L Zhao**, B Chen, K Li, HN Liang, L Yu, "3DStoryline: immersive visual storytelling," in Journal of Visualization, vol. 28, no. 2, pp. 463-480, Apr. 2025, doi:[10.1007/s12650-025-01058-5](https://doi.org/10.1007/s12650-025-01058-5).
- [5] **L. Zhao**, T. Isenberg, F. Xie, H.-N. Liang and L. Yu, "MeTACAST: Target- and Context-Aware Spatial Selection in VR," in IEEE Transactions on Visualization and Computer Graphics, vol. 30, no. 1, pp. 480-494, Jan. 2024, doi:[10.1109/TVCG.2023.3326517](https://doi.org/10.1109/TVCG.2023.3326517).
- [6] H. Yao, **L. Zhao**, H.-N. Liang, Y. Liu, Y.Li and L. Yu, "Exploring Embodied Asymmetric Two-Handed Interactions for Immersive Data Exploration," in CHI Conference on Human Factors in Computing Systems Extended Abstracts, issn: 979-8-4007-0331-7, May 2024, doi:[10.1145/3613905.365077](https://doi.org/10.1145/3613905.365077).
- [7] J. Li, **L. Zhao**, H.-N. Liang and L. Yu, "ImmerView: Adaptive Multi-View Layout for Immersive Situated Visualizations," in IEEE International Symposium on Mixed and Augmented Reality Adjunct, issn: 2771-1110, Oct. 2023, doi:[10.1109/ISMAR-Adjunct60411.2023.00030](https://doi.org/10.1109/ISMAR-Adjunct60411.2023.00030).
- [8] **L. Zhao**, N. Cao, S. He, H. -N. Liang and L. Yu, "L-WiM: Collaborative Exploration in Immersive Environments," in IEEE International Symposium on Mixed and Augmented Reality Adjunct, issn: 978-1-6654-5365-3, Oct. 2022, doi:[10.1109/ISMAR-Adjunct57072.2022.00031](https://doi.org/10.1109/ISMAR-Adjunct57072.2022.00031).
- [9] J. Zhang, M. Huang, **L. Zhao**, R. Yang, H.-N. Liang, J. Han, L. Wang and W. Sun., "Influence of Hand Representation Design on Presence and Embodiment in Virtual Environment," in International Symposium on Computational Intelligence and Design, issn: 978-1-7281-8446-3, Jan. 2021, doi: [10.1109/ISCID51228.2020.00088](https://doi.org/10.1109/ISCID51228.2020.00088).

### Peer-reviewed Posters

- [10] F. Xie, **L. Zhao**, N. Cao, S. He, and L. Yu. "L-WiM: Collaborative Exploration in Immersive Environments", in China Visualization and Visual Analytics Conference, Jul. 2023, **Best Poster Award**.
- [11] R. Zhou, F. Xie, Z. Yuan, **L. Zhao**, and L. Yu., "Selection, Annotation and Navigation for 3D Geological Data through Hybrid Interactions," in China Visualization and Visual Analytics Conference, Jul. 2022.

## AWARDS AND HONORS

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|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 07/2025 | <b>2025 XJTLU Honored PhD student.</b>                                                                                                                 |
| 05/2025 | <b>2025 XJTLU PhD Research Excellence Award.</b>                                                                                                       |
| 12/2023 | <b>Best Oral Presentation Award.</b> The First Wisdom Lake Postgraduate Researcher Development Conference& 2023 XJTLU Postgraduate Research Symposium. |
| 07/2023 | <b>Best Poster Award</b> (at ChinaVis2023) for "L-WiM:Collaborative Exploration in Immersive Environments".                                            |
| 07/2022 | <b>2<sup>nd</sup> prize</b> (at ChinaVis 2022 Data Visualization Competition) for "3D digital twin visualization scene design and construction".       |
| 12/2022 | <b>Excellent Poster Award.</b> XJTLU Postgraduate Research Symposium.                                                                                  |
| 12/2021 | <b>XJTLU Doctoral Scholarship.</b> XJTLU                                                                                                               |
| 06/2020 | <b>Full Scholarship.</b> XJTLU 3 <sup>rd</sup> Undergraduate.                                                                                          |
| 01/2019 | <b>Honorable Mention</b> (at Mathematical Contest in Modeling 2019) for "Send in the Drones: Developing an Aerial Disaster Relief Response System".    |
| 11/2018 | <b>Silver Medal</b> (at The University Physics Competition 2018) for "Sending a Light Sail Propelled Nanocraft to Alpha Centauri".                     |

## CONFERENCE TALKS

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|         |                                                                                                                                                      |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 03/2026 | <b>Paper Presentation.</b> "ScaleFree: Dynamic KDE for Multiscale Point Cloud Exploration in VR", IEEE VR 2026 (CCF A), Daegu, Republic of Korea.    |
| 05/2025 | <b>Invited talk.</b> Invited talk. "Outstanding PhD Student Forum", Shanghai Key Laboratory of Data Science, Fudan University, Shanghai, China.      |
| 10/2024 | <b>Paper Presentation.</b> "SpatialTouch: Exploring Spatial Data Visualizations in Cross-Reality", IEEE VIS 2024 (CCF A), California, United States. |
| 10/2023 | <b>Paper Presentation.</b> "MeTACAST: Target- and Context-Aware Spatial Selection in VR", IEEE VIS 2023 (CCF A), Melbourne, Australia.               |
| 10/2023 | <b>Doctoral Colloquium.</b> "Spatial selection techniques in the immersive environment", IEEE VIS 2023 (CCF A), Melbourne, Australia.                |
| 07/2023 | <b>Academic talk.</b> "Spatial selection techniques in VR", ChinaVis 2023, Chongqing, China.                                                         |
| 10/2023 | <b>Paper Presentation.</b> "ImmerView: Adaptive Multi-View Layout for Immersive Situated Visualizations", ISMAR 2023 (CCF B), Sydney, Australia.     |
| 10/2022 | <b>Paper Presentation.</b> "L-WiM: Collaborative Exploration in Immersive Environments", ISMAR 2022 (CCF B), Singapore.                              |
| 06/2020 | <b>Invited talk.</b> "Experience sharing on University Physics Competition/Mathematical Contest in Modeling", XJTLU,SB102                            |

## PROJECT EXPERIENCE

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| 01/2023–12/2026 | <b>NSFC General Programme.</b> Natural Interaction Techniques for Immersive Data Visualization. National Natural Science Foundation of China (Ongoing) (RMB 540,000). Assist to prepare the proposal.               |
| 06/2024–06/2025 | <b>State Key Lab Open Project.</b> Cross-reality Immersive Visualization and Exploration for Scientific Data. State Key Lab of CAD&CG, Zhejiang University. (Ongoing) (RMB 20,000). Assist to prepare the proposal. |
| 03/2025         | <b>NSFC Key Programme.</b> Large Model-driven Intelligent Visual Analytics and Interaction. National Natural Science Foundation of China (Submitted). Assist to prepare the proposal.                               |
| 03/2025         | <b>Outstanding Young-Scholar programme</b> in Jiangsu Science and Technology Programme. Cognitive                                                                                                                   |

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|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 | Enhancement and Intelligent Interaction Techniques in Immersive Environments. (Submitted). Assist to prepare the proposal.                                                                                                                                             |
| 09/2020–06/2023 | Collaborated with <b>SenseTime Group Inc.</b> Library Management and Book Retrieval System with Augmented Reality navigation techniques. Virtual Engineering Center, XJTLU. Attend and develop front-end and back-end computer vision and 3D reconstruction algorithm. |
| 06/2023–06/2024 | Collaborated with <b>biologists</b> . Visualization and Interaction Toolkit for Molecular Data in Mixed-reality Environment. Attend and develop visualization and interaction techniques for molecular data in MR. One paper has been published [2].                   |
| 06/2024–03/2025 | Collaborated with <b>The Fourth Affiliated Hospital of Soochow University</b> (Suzhou Dushu Lake Hospital). Deep Learning based Intelligent Repositioning Guidance System for Femur Fracture Surgery. Attend.                                                          |
| 11/2025–06/2026 | Collaborated with <b>XJTLU-graduate-school</b> . Developed an AI-driven research intelligence system to model faculty expertise at scale and support strategic academic decision-making through semantic analysis, knowledge profiling, and intelligent matching.      |

## ACADEMIC ACTIVITIES AND SERVICES

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### Conference Organization/Programme Committees

|         |                                                                                                                  |
|---------|------------------------------------------------------------------------------------------------------------------|
| 10/2024 | ISMAR workshop on " <a href="#">2nd Joint Workshop on Cross Reality</a> " (JWCR), <b>Workshop Co-organizer</b> . |
| 05/2025 | IEEE Visualization & Visual Analytics (VIS 2025), <b>Poster Programme Committee</b> .                            |
| 04/2026 | IEEE Visualization & Visual Analytics (VIS 2026), <b>Poster Programme Committee</b> .                            |
| 05/2026 | China Visualization and Visual Analytics Conference (ChinaVis 2026), <b>Programme Committee</b> .                |

### Student Volunteer

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|---------|---------------------------------------------------------------------|
| 10/2023 | IEEE Visualization & Visual Analytics (VIS 2023)                    |
| 10/2022 | International Symposium on Mixed and Augmented Reality (ISAMR 2022) |

### External Reviewing for Conference

|            |                                                                         |
|------------|-------------------------------------------------------------------------|
| IEEE VR    | IEEE Virtual Reality and 3D User Interfaces ( <b>CCF A</b> )            |
| CSCW       | Computer Supported Cooperative Work ( <b>CCF A</b> )                    |
| CHI        | CHI Conference on Human Factors in Computing Systems ( <b>CCF A</b> )   |
| ISMAR      | International Symposium on Mixed and Augmented Reality ( <b>CCF B</b> ) |
| EuroVis    | Eurographics Conference on Visualization ( <b>CCF B</b> )               |
| PacificVis | IEEE Pacific Visualization Symposium ( <b>CCF C</b> )                   |
| ISS        | ACM Interactive Surfaces and Spaces                                     |

### External Reviewing for Journal

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|------|----------------------------------------------------------------------------|
| TVCG | IEEE Transactions on Visualization and Computer Graphics ( <b>JCR Q1</b> ) |
| VI   | Visual Informatics ( <b>JCR Q3</b> )                                       |
| MCB  | Molecular & Cellular Biomechanics                                          |

## SELECTED MENTORSHIP EXPERIENCE

|            |                |                                                                           |
|------------|----------------|---------------------------------------------------------------------------|
| Jiayin Li  | Master student | MSc Human-Computer Interaction, XJTLU, published paper on ISMAR 2023 [6]. |
| Haonan Yao | Master student | MSc Human-Computer Interaction, XJTLU, published paper on ISMAR 2024 [5]. |

## TEACHING ASSISTANT

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|         |                                                                                         |
|---------|-----------------------------------------------------------------------------------------|
| 2021-24 | CPT405, Interactive Systems, graduate, XJTLU                                            |
| 2024-25 | CPT408/CPT306, Game Design, graduate, XJTLU                                             |
| 2021-23 | CPT401, Research Methods, graduate, XJTLU                                               |
| 2021-24 | CPT413, Information Visualization, graduate, XJTLU                                      |
| 2021-22 | CPT412, Human-Robot Interaction, graduate, XJTLU                                        |
| 2024-25 | CPT306, Game Design, undergraduate, XJTLU                                               |
| 2024-25 | CPT204, Advanced OO Programming, undergraduate, XJTLU                                   |
| 2022-23 | CPT105, Introduction to Programming in Java, undergraduate, XJTLU                       |
| 2022-23 | CPT102, Data Structures, undergraduate, XJTLU                                           |
| 2022-23 | CPT106, C++ Programming and Software Eng. II, undergraduate, XJTLU                      |
| 2022-23 | Professional Development Programme (Coding with python), undergraduate, XJTLU (Taicang) |